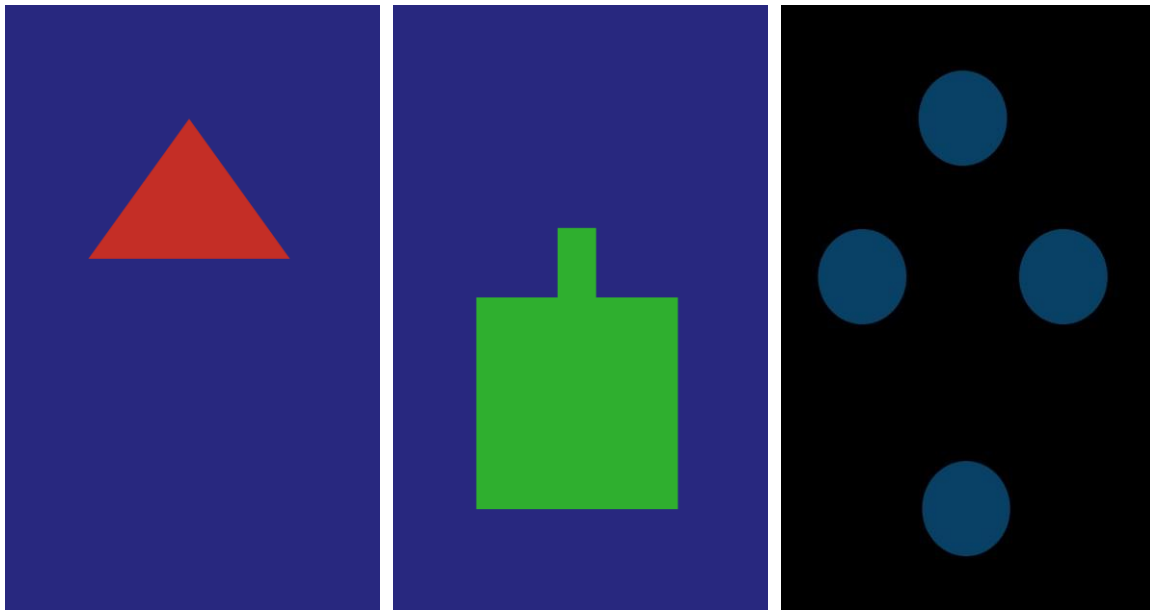


Course	Digital Image Processing
Professor	Kwang Nam Choi
Date	September 17 th , 2026
Submission and Q&A board	https://eclass3.cau.ac.kr/

Digital Image Processing Team Project Specification

Sample Image



squid_head

squid_body

squid_points

Requirement

Implementation of a program to generate squid game images using **composite function**.

Flow Sequence : 1. Add squid_head and squid_body. 2. Subtract squid_points from the result.

1. Team project assignment

Each of student assign team through e-class team project menu.

※ Team should be made up of 2 students.

2. Related references

CxImage Library : <https://www.codeproject.com/Articles/1300/CxImage>

Source Images – Head, Body, Points

3. Submission

- 1) Submit the source codes of your project that can be compiled in IDE(Visual Studio).
 - Submit folder: e-Class / Team Project 01
 - 2) Submit a final report(word) that includes the contents below.
 - ① Outline of program
 - ② Design of program
 - ③ The spending time
 - ④ Process of project and record of project meeting (Picture, Kakao Talk capture, or etc.)
 - ⑤ Each team member's thoughts on this project
 - ⑥ Etc.
- Submit folder: e-Class / Team Project 01
 - File name: TeamNumber_TeamName.zip (ex: *Team01_DIP.zip*)
 - **Deadline : 10:00 PM, Thur. October 29th (Week 9), 2026**

4. Notifications

- Test image should be in the same folder.
- Input: 3 images ('squid_head.png', 'squid_body.png', 'squid_points.png') at the same time.
- Shape, location and size of input images can be changed slightly.
- Questions and answers utilize the class or the bulletin board below.
- In the result window, the overlapping part of the figure should have a different color from the existing figures.
- Please submit report, result image and source file as compressed file (*.zip).
- Except bellow the functions in CXImage, do not use the any pre-implemented function in libraries.
ex)
 - Allowed: GetHeight(), GetWidth(), GetPixelColor(), SetPixelColor() ...
 - Denied: RGB2GRAY ...
- You must submit the complete Visual Studio project source files (.sln, .vcxproj, .cpp, .h, etc.).
- The submitted project will be verified by opening the .sln file and building and running it in Visual Studio.
- The project must be buildable and executable in Visual Studio without any modification.
- The .sln file name must follow the format teamNumber_teamName.sln (e.g., team1_vim.sln).
- **The result image must be included in the final report. Failure to include the result image will result in a penalty.**

5. Related references

OpenCV Library : <http://opencv.org/>

OpenCVKorea : <http://www.opencv.co.kr>

- ※ These websites provide the tutorials, practice for image I/O and the documentations like user's guide and reference manual for Digital Image Processing with OpenCV library.
- ※ However, Using the pre-implemented functions in OpenCV library which used to detect object, such as binarization, grayscaling, edge detection, labeling, contour extraction and etc, will be penalized on grade.

6. Submission

- 1) Submit the source codes of your project that can be compiled in IDE(Visual Studio).
 - Submit folder: e-Class / Team Project 01_extra
- 2) Submit a final report(word) that includes the contents below.
 - ① Outline of program
 - ② Design of program
 - ③ The spending time
 - ④ Process of project and record of project meeting (Picture, Kakao Talk capture, or etc.)
 - ⑤ Each team member's thoughts on this project
 - ⑥ Etc.
 - Submit folder: e-Class / Team Project 01_extra
 - File name: TeamNumber_TeamName_extra.zip (ex: *Team01_VIM_extra.zip*)
 - **Deadline : 10:00 PM, Thur. October 29th (Week 9), 2026**